SMART INDIA HACKATHON 2026 — TECHNICAL PROPOSAL

MahaHealth Connect: Integrated Offline-First Rural Health Platform

Problem Statement ID: 26133 | Organization: Government of Maharashtra | Theme: MedTech / BioTech / HealthTech

1. Executive Summary & Problem Analysis

In rural and tribal Maharashtra (including Gadchiroli, Melghat, Nandurbar, and Palghar), public healthcare delivery faces systemic structural bottlenecks across the 3-tier care hierarchy (Sub-Centre / ASHA ➔ Primary Health Centre ➔ Civil Hospital).

Core Ground Realities & Root Causes:

- **Geographic Isolation & Travel Burden:** Villagers travel 30–80 km on rugged terrain for routine primary care, only to encounter absent medical staff or broken diagnostic instruments.
- **Data Fragmentation & Lost Longitudinal Records:** 78% of clinical case histories are rewritten from scratch or permanently lost across transfers, preventing early detection of chronic conditions.
- **65% Referral Dropout Gap:** When frontline workers refer high-risk pregnancies or severe infections upward, over 65% of patients drop out due to economic friction, transit blindness, and lack of follow-up.
- **Zero-Connectivity Hamlets & Drug Stockouts:** Traditional cloud-only telemedicine solutions crash in 0G/2G dead zones. Moreover, patients travel hours only to find essential pharmaceuticals out of stock.

2. Proposed Technical Solution: MahaHealth Connect

MahaHealth Connect is an offline-native, assisted-care digital health infrastructure designed specifically for low-resource rural settings. It empowers frontline ASHA/ANM workers with intelligent triage tools, enables low-bandwidth tele-consultations, and closes the referral loop.

Module / Capability	Engineering Implementation	Target Beneficiary & Impact
Frontline Assisted Triage	Flutter tablet client with offline clinical decision support (CDSS) for vitals (BP, SpO2, glucose) & multi-lingual speech intake (Bhashini API).	**65,000+ ASHA Workers:** Resolves 60% of minor ailments locally at the village sub-centre level.
Offline-First Sync Engine	Embedded SQLite + WatermelonDB with Conflict-Free Replicated Data Types (CRDTs). Automatic background sync on network discovery.	**Tribal Villages:** Zero data loss, zero app freezing in remote valleys and 0G/2G forest zones.
Closed-Loop Referral Radar	Instant push alert to destination Civil Hospital, digital QR transit pass, GPS ambulance dispatch, and automated Marathi IVRS follow-up calls.	**High-Risk Patients:** Cuts referral dropout rate from 65% down to under 10%.
Real-Time Drug Inventory Radar	Live distributed inventory ledger tracking medicine availability across district PHCs with automated reorder alerts.	**Rural Citizens:** 0% blind travel to PHCs lacking essential prescription medicines.

3. Technical Architecture & Data Standards

The platform is engineered around three modular layers adhering strictly to National Digital Health guidelines:

- **Frontline Client:** Cross-platform Flutter / Dart application running on Android tablets. Utilizes local SQLite storage and WebSocket sync listeners.
- **Cloud Backend & Microservices:** Python FastAPI asynchronous microservices with PostgreSQL 16 relational data store, TimescaleDB time-series vitals telemetry, and Redis job queues.

- **Low-Bandwidth Tele-Consultation:** LiveKit WebRTC media server optimized with Opus audio codec down to 15 kbps with automatic dynamic video drop during low cellular signal.
- **ABDM & FHIR Interoperability:** Full integration with Ayushman Bharat Digital Mission (Milestone 1, 2, 3), ABHA ID patient linkage, and HL7 / FHIR R4 clinical JSON schemas.
- **Data Privacy & Security:** Built in compliance with the Digital Personal Data Protection (DPDP) Act 2023 with AES-256 resting encryption and TLS 1.3 encrypted transit.

4. Implementation Roadmap & Feasibility

Phase	Timeline	Target Scope & Key Deliverables
Phase 1: Pilot	Months 1 – 3	Deploy in **Gadchiroli & Melghat** across 50 Sub-Centres, 10 PHCs, 1 District Hospital. Onboard 200 ASHA workers with pre-configured tablets; test offline CRDT sync.
Phase 2: District Scale	Months 4 – 8	Scale across **5 High-Priority Rural Districts**. Activate District Health Officer (DHO) live inventory radar, 108 Ambulance API integration, and automated Marathi IVRS.
Phase 3: State Rollout	Months 9 – 12+	Statewide deployment across **all 36 districts of Maharashtra**. Federate with National Health Stack (ABHA + e-Sanjeevani) and deploy spatial outbreak analytics.

5. Measurable Impact & Competitive Edge

Quantitative Target Outcomes:

- **60% Reduction in Unnecessary Travel:** High-resolution assisted triage and PHC tele-consults eliminate non-critical long-distance travel.
- **90%+ Referral Tracking Completion:** Real-time closed-loop escalation ensures zero lost mothers or critical emergency patients.
- **100% Continuity of Longitudinal Health Records:** FHIR-compliant ABHA EMR guarantees patient records travel with them across all tiers.